Contents

- Exercise 1 %% %Basic matrix and vector operations
- Exerise 2 %% %Plotting functions through Matlab
- Exercise 3 %% %Plotting car path from Initial Position to Target


```
%ECE8 Lab 1
```

Exercise 1 %% %Basic matrix and vector operations

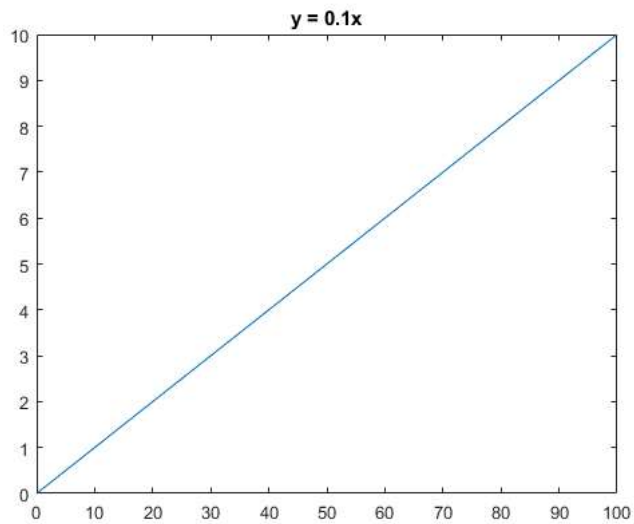
```
%Define all variables for later use, followed with semi-colons so they
% don't show up in the terminal.
A = [8 1 6; 3 5 7; 4 9 2];
B = [29 44 86; 1 66 37; 84 78 5];
C = [7 3 9; 6 3 5; 7 6 5];
x = [2; 7; 9];
u = [9; 38; 45];

%Execute parts a, b, and c with the defined variables
%Part (a)
('Exercise 1')
PartA = A*B + C

%Part (b)
PartB = A*x + B*u

%Part (c)
PartC = (1/3)*A*(x + u)
```

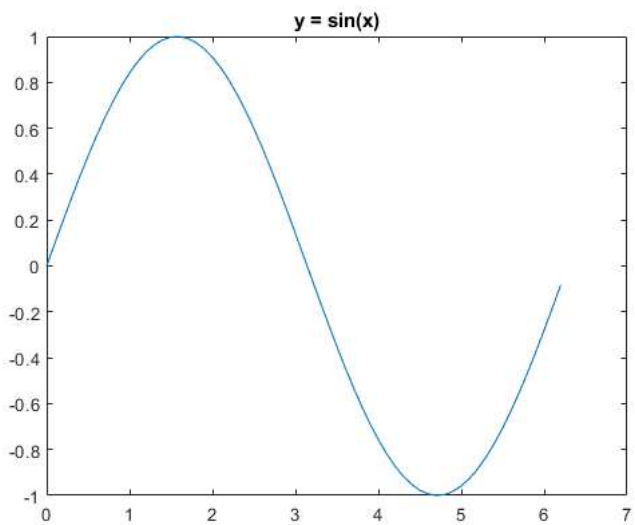
```
ans =

    'Exercise 1'


PartA =

         744         889         764
         686        1011         483
         300         932         692


PartB =

        5880
        4286
        4034


PartC =

  152.3333
  212.0000
  185.6667
```

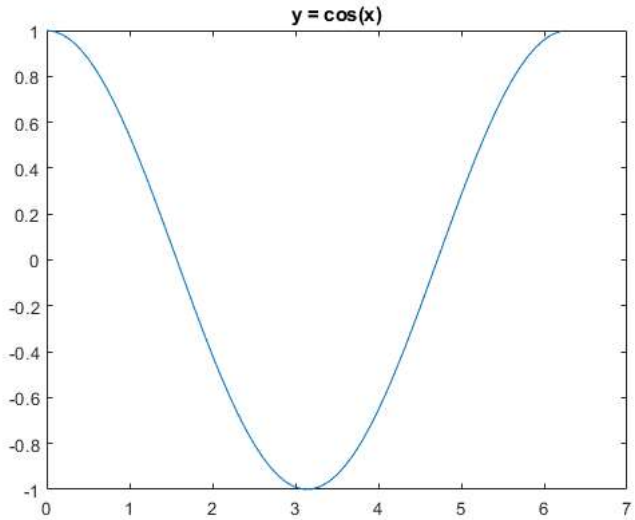
Exerise 2 %% %Plotting functions through Matlab

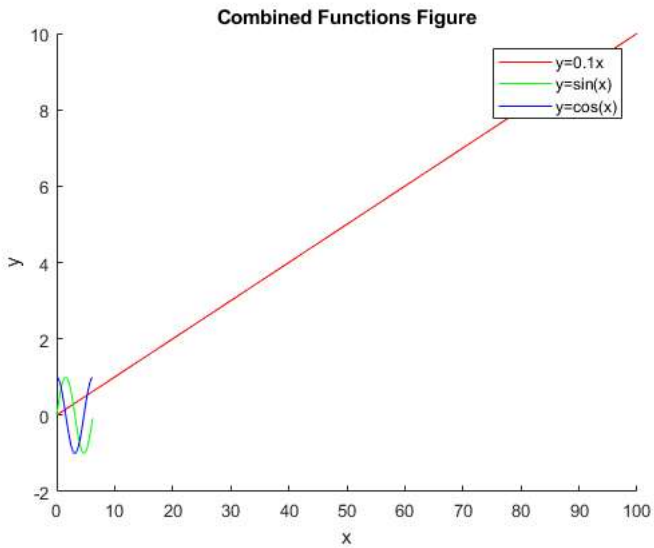
```
% First, define all the variables to use later
%Part (a)
x1 = 0:2:100;
y1 = 0.1*x1;

%Part (b)
x2 = 0:0.1:2 * pi;
y2 = sin(x2);

%Part (c)
x3 = 0:0.1:2 * pi;
y3 = cos(x3);

% Now, for the individual plots
% I will specify each plot as a seperate figure to have them all show up
% seperately
figure(1)           % First plot, y = 0.1x %
plot(x1, y1)
title('y = 0.1x')

figure(2)           % Second plot, y = sin(x) %
plot(x2, y2)
title('y = sin(x)')
```

```
figure(3)           % Third plot, y = cos(x) %
plot(x3, y3)
title('y = cos(x)')

% Now for the combined plot with all functions %
figure(4)
hold on
plot(x1, y1, 'r', 'DisplayName', 'y=0.1x')   %I put some labeling on each
plot(x2, y2, 'g', 'DisplayName', 'y=sin(x)') %line to make it more readable
plot(x3, y3, 'b', 'DisplayName', 'y=cos(x)')
hold off

title('Combined Functions Figure')           %Label everything and
xlabel('x')                                  %create a legend
ylabel('y')
legend
```





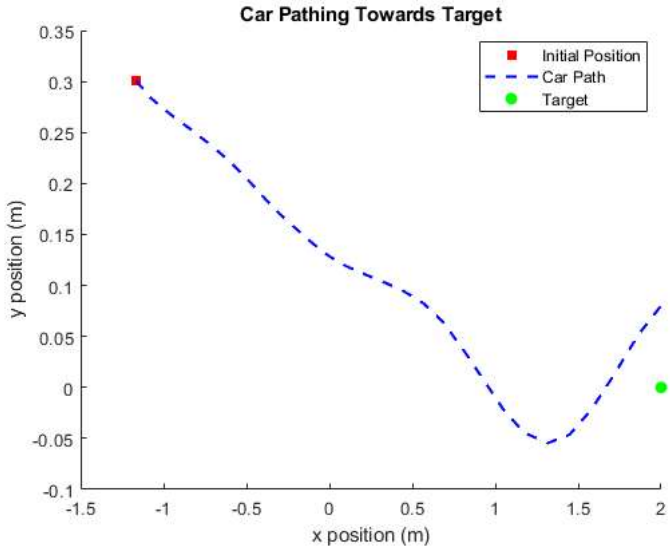




Exercise 3 %% %Plotting car path from Initial Position to Target

```
load('Lab1_Exercise3.mat')  %First load the data

figure(5)
hold on                     %Prepare the figure

% Initial Location Marker
plot(xv(1), yv(1), ...
    'Color', 'r', ...
    'Marker', 's', ...
    'MarkerFaceColor', 'r', ...
    'LineStyle', 'none', ...
    'DisplayName', 'Initial Position');      % a lot to read, but basically takes the first piece of data for x and y, plots a marker as a red square, and labels it

% Plotting Car Path
plot(xv, yv, ...
    'Color', 'b', ...
    'LineStyle', '--', ...
    'LineWidth', 1.5, ...
    'DisplayName', 'Car Path');              % Similar story. Creates a blue dashed line for the car's path, makes it a certain width, and labels it as "Car Path"

% Plotting End Marker
plot(2, 0, ...
    'Color', 'g', ...
    'Marker', 'o', ...
    'MarkerFaceColor', 'g', ...
    'LineStyle', 'none', ...
    'DisplayName', 'Target');                % Almost Identical to the initial location marker, but we make it a green circle and call it "Target"

% Labels
xlabel('x position (m)')                %Label all the axis and legend to make everything more readable
ylabel('y position (m)')
title('Car Pathing Towards Target')
```

```
legend('Location', 'best')
hold off
```



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